

Automatic Query-based Personalized Summarization that uses Pseudo Relevance Feedback with NMF

Sun Park

Advanced Graduate Education Center
of Jeonbuk for Electronics and
Information Technology-BK21
Chonbuk National University,
South KOREA
+82-63-270-4341
sunbak@jbnu.ac.kr

Dong Un An

Division of Electronic & Information
Engineering
Chonbuk National University,
South KOREA
+82-63-270-2416
duan@jbnu.ac.kr

ABSTRACT

This paper proposes a new automatic query-based personalized summarization using Pseudo Relevance Feedback (PRF) with Non-negative Matrix Factorization (NMF) to extract meaningful sentences from to retrieve documents in Internet. The proposed method can improve the quality of personalized summaries because the inherent semantics of the documents are well reflected by using the semantic features calculated by NMF and the sentences most relevant to the given query are extracted efficiently by using PRF with NMF. Besides, it can reduce the semantic gap between the low level search result and high level user's perception by means of PRF with semantic features. The experimental results demonstrate that the proposed method achieves better performance than the other methods.

Categories and Subject Descriptors

H.3.1 [Information Storage and Retrieval]: Content Analysis and Indexing - abstracting methods.

General Terms

Algorithms, Experimentation.

Keywords

Personalized document summarization, pseudo relevance feedback, NMF, semantic features.

1. INTRODUCTION

The personalized summarization is the process of summarization that preserves the specific information that is relevant for a given user profile rather than information that truly summarizes the content of the news item [1]. If the summary is personalized according to user interests, the user can save time not only in deciding whether it is interesting or not, but also in finding the information without having to read the full text [1]. The automatic document summarization is the process of reducing the sizes of documents while maintaining their basic outlines. That is, it should distill the most important information from the document. It can be either generic summaries or query based summaries. A generic summary distills an overall sense of the documents' contents whereas a query based summary only distills the contents of the document relevant to the user's query [9].

Due to the increase in the amount of accessible text documents on the Internet, the necessity for automatic personalized document summarization has also increased. However, the automatic personalized summarization methods by means of machine are poor because the semantic gap between a summary by means of high-level users' requirement and a summary of low-level machine exist. In order to overcome this problem with document summarization methods, recent studies that have used the User's Feedback (UF) [2, 3, 5, 9] and NMF [10, 11, 12, 13, 14, 15, 16, 17, 18, 19] tried to reduce the semantic gap between the low-level features and high-level concepts.

Generally, UF can be either Relevance Feedback (RF) or Pseudo Relevance Feedback (PRF). As one of the RF in information retrieval, a query reformulation method is a well known approach. However, it needs an intervention of a user for a relevance judgment on documents and the bias of query expansion can make the semantic gap problem get worse. The PRF can provide an automatic relevance judgment on sentences without user's intervention but it may get biased query during a query expansion process [2, 3, 9, 20]. To solve this problem in a document summarization fields, it use *query splitting* [3] and NMF [16, 17]. The *query splitting* decomposes the initial query into several pieces of queries by constructing a new feedback query for each relevant sentence [3]. The NMF can represent individual object as the non-negative linear combination of part information extracted from a large volume of objects. The NMF have a great power to easily extract semantic features representing the inherent structure of data objects. The factorization result of NMF has better semantic interpretation [6, 7, 24].

In this paper, we propose a new method that uses Pseudo Relevance Feedback with NMF to summarize a personalized summarization with regard to a given query. The propose method has the following advantages: First, it can select sentences covering the major topics of the search results by using the semantic features by means of NMF. Second, it can improve the quality of personalized summarization since extracting sentences to reflect the inherent semantics of a search result by generic and query-based summarizations. Third, it can select sentences that are highly relevant to a given query because it can expand the query with respect to user interesting by using PRF with NMF.

The rest of the paper is organized as follows: Section 3, we describe the query-based document summarization methods and the NMF algorithm in detail. In Section 4, the new personalized

summarization method is introduced. Section 5 shows the evaluation and experimental results. Finally, we conclude in Section 5.

2. Related Works

2.1 Personalized document summarization

The previous studies for query-based summarization are as follows: Diaz and Gervas proposed automatic personalized summarization using combination of generic and query-based methods. Their generic summarization methods combine position method with thematic word method. Their personalized method is to select those sentences of a document that are most relevant to a given user model [1]. Ko et al. [5] proposed web snippet generation method using PRF and statistical query expansion. Han et al. proposed a text summarization using relevance feedback with query splitting. Their method can alleviate the problem that feedback gets biased query during a query expansion process by splitting the initial query into several pieces. If it has insufficient information for query splitting, it may produce poor summaries of documents [3].

Park et al. proposed a query-based summarization method using NMF [10]. This method extracts sentences using the similarity between a query and semantic features extracted by NMF. Park et al. also proposed a multi-document summarization method based on clustering using NMF [11]. This method clusters the sentences and extracts sentences using the cosine similarity measure between a topic and semantic features. Park et al. proposed a document summarization using non-negative matrix factorization and relevance feedback [16].

Pack et al. proposed automatic personalized summarization using non-negative matrix factorization and relevance measure [13]. Park proposed personalized summarization using NMF and Relevance Feedback [17]. Park proposed personalized document summarization using semantic features by NMF [14]. Park proposed personalized summarization agent using sentences ranking with respect to semantic features by NMF [15, 19].

Tombros and Sanderson proposed a query biased summaries. Their method generates user adapted summaries by combining title score, heading score, term occurrence information, and query score [22].

Sanderson proposed an accurate user directed summarization using best passage operator and query expansion from Local Context Analysis (LCA). His method selects the passage most relevant to the query, previously expanded with the words that occur most frequently in the context in which the words of the query appear in the first document retrieved [21].

Varadarajan and Hristidis propose query specific document summarization method by identifying the most query relevant fragments and combining them using the semantic associations within the document [23].

2.2 Non-negative Matrix Factorization

In this paper, we define the matrix notation as follows: Let X_{*j} be j 'th column vector of matrix X , X_{i*} be i 'th row vector, and X_{ij} be the element of i 'th row and j 'th column.

Non-negative Matrix Factorization (NMF) is to decompose a given $m \times n$ matrix A into a non-negative semantic feature matrix W and a non-negative semantic variable matrix H as shown in Equation (1).

$$A \approx WH \quad (1)$$

where W is a $m \times r$ non-negative matrix and H is a $r \times n$ non-negative matrix. Usually r is chosen to be smaller than m or n , so that the total sizes of W and H are smaller than that of the original matrix A .

The objective function is used minimizing the Euclidean distance between each column of A and its approximation $\tilde{A} = WH$, which was proposed in Lee and Seung [6, 7, 24]. As an objective function, the Frobenius norm is used [24]:

$$\Theta_E(W, H) \equiv \|A - WH\|_F^2 \equiv \sum_{j=1}^m \sum_{i=1}^n \left(X_{ji} - \sum_{l=1}^r W_{jl} H_{li} \right)^2 \quad (2)$$

Updating W and H is kept until $\Theta_E(W, H)$ converges under the predefined threshold or exceeds the number of repetition. The update rules are as follows:

$$H_{\alpha\mu} \leftarrow H_{\alpha\mu} \frac{(W^T A)_{\alpha\mu}}{(W^T WH)_{\alpha\mu}}, \quad W_{i\alpha} \leftarrow W_{i\alpha} \frac{(AH^T)_{i\alpha}}{(WHH^T)_{i\alpha}} \quad (3)$$

A column vector corresponding to j 'th sentence A_{*j} can be represented as a linear combination of semantic feature vectors W_{*l} and semantic variable H_{lj} as follows:

$$A_{*j} = \sum_{l=1}^r H_{lj} W_{*l} \quad (4)$$

where W_{*l} is the l 'th column vector of W .

The powers of the two non-negative matrices W and H are described as follows: All semantic variables (H_{lj}) are used to represent each sentence. W and H are represented sparsely. Intuitively, it makes more sense for each sentence to be associated with some small subset of a large array of topics (W_{*l}), rather than just one topic or all the topics. In each semantic feature (W_{*l}), the NMF has grouped together semantically related terms. In addition to grouping semantically related terms together into semantic features, the NMF uses context to differentiate between multiple meanings of the same term [6].

Example 1) We illustrate an example using NMF [6, 7, 24]. In NMF, let r be 2, the number of repetitions 50, and the tolerance 0.001. When the initial elements of the W and H matrices are 0.5, the non-negative matrix A is decomposed into two non-negative matrices, W and H , as shown in Figure 1(c). Figure 1(d) shows an example of sentence representation using NMF. The column vector A_{*j} corresponding to the first sentence is represented as a linear combination of the semantic feature vectors W_{*l} and the semantic variable column vector H_{*1} .

$$A \approx W \times H$$

$$\begin{bmatrix} 1 & 1 & 4 & 5 \\ 0 & 0 & 8 & 0 \\ 10 & 0 & 0 & 12 \end{bmatrix} \approx \begin{bmatrix} 0.613 & 1.154 \\ 2.169 & 0 \\ 0 & 2.193 \\ 0.112 & 0.010 \end{bmatrix} \times \begin{bmatrix} 4.380 & 0.096 & 0 & 50263 \\ 0 & 0.177 & 3.608 & 0.334 \end{bmatrix}$$

(a) Result of NMF decomposition

$$\begin{bmatrix} 1 \\ 0 \\ 10 \end{bmatrix} \approx 4.380 \times \begin{bmatrix} 0.613 \\ 2.169 \\ 0 \\ 0.112 \end{bmatrix} + 0 \times \begin{bmatrix} 1.154 \\ 0 \\ 2.193 \\ 0.010 \end{bmatrix}$$

$$A_{s1} \quad H_{11} \quad W_{s1} \quad H_{21} \quad W_{s2}$$

(b) Example of sentence representation using semantic features and semantic variables by NMF

Figure 1. Example of sentence representation using semantic features and their weights by NMF

Example 2) Table 1 shows a part of extracted sentences from a document related with the topic “Native American Reservation System - pros and cons” in *DUC* 2006. The term-by-sentence matrix A is generated by preprocessing a set of sentences in Table 1. The matrix A composes of 541 terms and 54 sentences. Table 2 illustrates the cases of applying the *NMF* to the matrix A .

Table 1. A part of sentences related with topic “Native American Reservation System - pros and cons” in *DUC* 2006

The number of sentences	Sentences
...	...
S20	Indians working on reservations avoid state income taxes but pay federal income and Social Security taxes. They can vote in local, state and federal elections _ a right that didn't exist for them in New Mexico, for instance, until 1948. But a non-Indian resident of a reservation has no say in tribal government.
...	...
S37	Neighbors to a new prison in south-central Missouri battled mightily, but unsuccessfully, to stop the \$73 million project. But the Iowa tribe of Oklahoma was able to halt construction last month. The tribe said excavation could upset the remains of their ancestors _ some of the 70,000 American Indians forced to move cross-country by the Indian Removal Act of 1830.
...	...
S42	Although Utah hasn't explicitly conceded that the tribe can legally store the waste, the governor and the Legislature have chosen to point their efforts at the transportation of radioactive material over public lands rather than challenge outright what the Goshutes can do on their reservation.

Table 2 illustrates 10 semantic feature vectors, $W_{s1}, \dots, W_{s10}$, obtained from the *NMF* decomposition of the matrix A , weight values, $H_{1,20}, \dots, H_{10,20}$, of semantic feature vectors with respect to the sentence S20, the original sentence vector, and the sentence vector calculated from the weight values and the semantic feature vectors.

There are zero values and no negative values in Table 2. That is, semantic feature vectors obtained by using *NMF* are sparse so that *NMF* can obtain semantic features having a small range of semantics. This indicates that a method that uses *NMF* has better power to identify subtopics of documents than do methods that

use decomposition approaches, such as Principal Components Analysis (PCA) and Vector Quantization (VQ). Besides, the semantic feature vectors in Figure 1 intuitively make more sense because *NMF* represents a sentence as the linear combination of a few intuitive semantic feature vectors having non-negative values [6].

Table 2. The representation of sentence by means of *NMF*

Term		Semantic feature								Sentence S20	
		W_{s_1}	W_{s_2}	W_{s_3}	W_{s_4}	W_{s_5}		W_{s_9}	$W_{s_{10}}$	Original	$\sum_{i=1}^{10} H_{i20} W_{s_i}$
1	crow	0.0155	0	0	0.1924	0.2509	...	0	1.1328	0	0.22624
...	...	...	...	...	...	...	...	...	...	...	...
40	horizon	0	0	0	0.154	0	...	0	0	0	0.00001
41	south	1.4191	3.6039	3.3799	2.6458	2.4167	...	2.577	0.0134	2	2.17768
42	central	0	0	0	0.1619	1.3247	...	1.3362	1.7151	1	1.1928
...	...	...	...	...	...	...	...	...	...	...	...
80	case	0	0	0	0	0.0649	...	1.245	0.0037	0	0.05902
81	few	0	0	0	0	1.1455	...			1	1.03164
82	find	0	0	0	0	0.082	...	0.0428	0.059	0	0.07452
...	...	...	...	...	...	...	...	...	...	...	...
170	silver	0.0001	0	0.6447	0.0001	0	...	0	0	0	0.00928
...	...	...	...	...	...	...	...	...	...	...	...
541	http	0	0	0	0	0	...	0	0	0	0.00099
Weight H_{ij}		0.0009	0	0.0001	0.0001	0.9002	...	...	0.000		

3. Automatic Personalized Summarization Method

In this paper, we propose an automatic query-based personalized document summarization using PRF with *NMF*. The proposed method consists of the preprocessing phase, the sentence extraction phase, and personalized summary generation phase. We will give a full explanation of three phases as Figure 2.

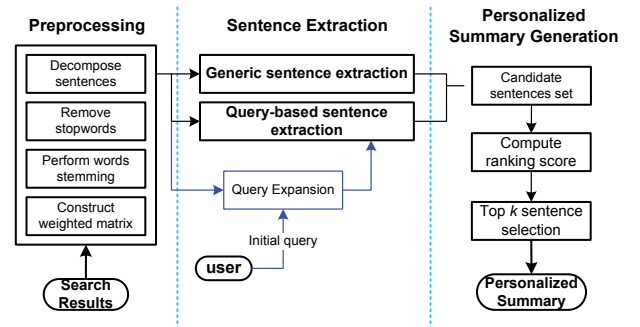


Figure 2. Personalized Summarization method

3.1 Preprocessing phase

In the preprocessing phase, after given search results are decomposed into individual sentences, stop-words are removed and word stemming is performed. Then the weighted term-frequency vector is constructed for each sentence in search results using Equation (5) [4, 20]. Let A be $m \times n$ matrix, where m is the number of terms and n is the number of sentences in the whole search results. Let element A_{ji} be the weighted term-frequency of term j in sentence i .

$$A_{ji} = L_{ji} \cdot G(j) \quad (5)$$

where L_{ji} is the local weight(term frequency) for term j in the sentence i , and $G(j)$ is the global weight(inverse document frequency) for term j in the whole search results [4, 20]. That is,

$$G(j) = \log(N/N(j)) \quad (6)$$

where N is the total number of sentences in the whole search results, and $N(j)$ is the number of sentences that contain term j .

3.2 Sentence extraction phase

Sentence extraction phase consists of generic summarization, PRF, and query-based summarization.

3.2.1 Generic summarization

We modify our previous generic summarization method using non-negative semantic variable by NMF [8, 12]. In the generic summarization, the candidate sentence set is constructed by using the generic relevance weight. The preprocessing phase and NMF are performed. Sentences with respect to the high generic relevance weight are selected. The relevance score with respect to the selected sentences is calculated by using Equation (9). The selected sentences are added to the candidate sentence set. We define the generic relevance weight $Grweight()$ as follows:

$$Grweight(H_{i*}) = \frac{\sum_{j=1}^r H_{ji} \times \frac{\sum_{q=1}^n H_{jq}}{\sum_{p=1}^r \sum_{q=1}^n H_{pq}}}{\sum_{p=1}^r \sum_{q=1}^n H_{pq}} \quad (7)$$

It means how much the sentence reflects major topic which is represented as semantic features.

$$sim(W_{*i}, Q) = \frac{W_{*i} \cdot Q}{|W_{*i}| \times |Q|} = \frac{\sum_{j=1}^m W_{ji} \times q_j}{\sqrt{\sum_{j=1}^m W_{ji}^2} \times \sqrt{\sum_{j=1}^m q_j^2}} \quad (8)$$

where a query vector $Q = (q_1, q_2, \dots, q_m)$. q_j denotes the j 'th term frequency of the query, while m denotes the number of terms.

The relevance score means how much the selected sentence reflects user interesting which are represented as the generic relevance weight of a sentence. Relevance score r_i is defined as follows.

$$r_i = Grweight(H_{i*}) \times sim(W_{*i}, Q) \quad (9)$$

where r_i is a relevance score of i 'th sentence.

The proposed algorithm for generic document summarization is as follows:

1. Perform the preprocessing phase.
2. Perform the NMF
3. for each sentence i
 Calculate $Grweight(H_{i*})$
4. Select k sentences with highest generic relevance weight values
5. Calculate relevance score r
6. Put the selected sentences into candidate sentence set

Pseudo relevance feedback

The PRF phase is described as follows: In the first step, for a given initial query, relevance sentences are extracted by the cosine similarity between the initial query and a sentence vector in documents by using Equation (8). The top k ranked sentences are then selected if they have similarly high values. The query expansion is then performed by using the extracted top k ranked sentences.

In the second step, the initial query is expanded by using each relevant sentence. The following query expansion method is used:

Query Point Movement:

$$Q^{new} = Q + \frac{\sum_{t=1}^k w_t \times A_{*i_t}}{\sum_{t=1}^k w_t}, w_t = sim(Q, A_{*i_t}) \quad (10)$$

where Q^{new} is the new expanded query vector of current query Q , A_{*i_t} is the t 'th sentence in the relevant sentences, w_t is the weight and is calculated by using the cosine similarity between the current query Q and A_{*i_t} .

3.2.2 Query-based summarization

We modify our previous query-based summarization method using non-negative semantic features and non-negative semantic variable by using NMF [10, 11].

The personalized summarization is described as follows. To evaluate the degree of similarity of the semantic feature vector W_{*i} with regard to the query Q^{new} as the correlation between the vector W_{*i} and Q^{new} used Equation (8). To select those sentences of a document that is most relevant to a given query. The personalized score with respect to the selected sentences is calculated by using Equation (11). The selected sentences are added to the candidate sentence set.

$$e_i = \sum_{j=1}^n H_{ij} \times sim(W_{*i}, Q) \quad (11)$$

The personalized score means how much the selected sentence reflects user interesting which are represented as the semantic features.

We use the following personalized summarization algorithm:

1. Perform the preprocessing phase.
2. Perform the NMF
3. Select a column vector W_{*p} of matrix W whose similarity to the query Q^{new} is the largest.
4. Select the sentence corresponding to the largest index value of the row vector H_{p*} .
5. Include the selected sentence in the candidate sentences set.
6. Compute the personalized score e_i .
7. If the number of selected sentences reaches the predefined number of selected number for each expanded query k , then stop the algorithm. Otherwise go to step 3 to find the next most similar column vector excluding W_{*p} .

In step 3, the fact that the similarity between W_{*p} and the query is largest means the p 'th semantic feature vector W_{*p} is the most relevant feature to the query. In step 4, it selects the sentence that has the largest weight for the most relevant semantic feature.

3.2.3 Personalized summary generation phase

Personalized summary generation phase using combination of generic and query-based summarization extracts top k ranked sentences from the candidate sentences set for personalized summary. This phase is described as follows: The relevance scores and the personalized scores are normalized. Then the ranking score of the candidate sentences is calculated by using Equation (12).

$$rs_i = \frac{\alpha \cdot r_i + \beta \cdot e_i}{\alpha + \beta} \quad (12)$$

where rs_i is a ranking score of i 'th sentence, parameters α and β is adjustment between the generic summarization and the query-based summarization. The value of parameters decides the degree of personalization of the resulting summaries: if β is 0, resulting summaries will be generic, if β is larger than 0, resulting summaries have a degree of personalization. In this paper, let α and β be 0.5.

4. Experimental Results

As an experimental data, we used Yahoo-Korea News¹. We gave a query to retrieve news documents from Yahoo-Korea News. Three independent evaluators were employed to manually create summarization on the 500 document from the retrieved Yahoo Korea news documents with respect to 20 queries. Each document in Yahoo Korea news has an average of 15.1 sentences selected by evaluators. Table 3 provides the particulars of the evaluation data set. The retrieved news documents are preprocessed using HAM (Hangul Analysis Module) which is a Korean language analysis tool based on Morpheme analyzer [4].

Table 3. Particulars of the Evaluation Data Corpus

Document attributes	Yahoo-Korea News
Number of docs	500
Number of docs with more than 10 sentences	321
Avg sentences / doc	15.1
Min sentences / doc	10
Max sentences / doc	103

In this paper, we used the recall (R), precision (P), and F -measure to evaluate the performance of the proposed method. Let S_{man} , S_{sum} be the set of sentences selected by the human evaluators, and the summarizer, respectively. The standard definitions of recall (R), precision (P), and F -measure are defined as follows [10]:

$$R = \frac{|S_{man} \cap S_{sum}|}{|S_{sum}|} \quad (13)$$

$$P = \frac{|S_{man} \cap S_{sum}|}{|S_{man}|} \quad (14)$$

$$F = \frac{2RP}{R + P} \quad (15)$$

We implemented 8 different summarization methods such as CRNMF, UPS, PNMF, PSFV, TNMF, RFNMF, PGNMF, and PRFNMF. The UPS [1] denotes Diaz's method using user-model based personalized summarization. The PNMF [13] denotes Park's method using the Relevance Measure and NMF for personalized summary. The PSFV [14] denotes Park's method using NMF for personalized summary. The TNMF [15] denotes Park's method using generic weighted NMF for personalized summary. The PGNMF [19] denotes Park's method using sentence ranking and summaries generation by using weighted NMF for personalized summary. The RFNMF [17] denote Park's method using relevance feedback with NMF for personalized summary. The CRNMF [18] denote Park's method using NMF with cluster refinement for personalized summary. The PRFNMF denote our proposed method using PRF with NMF.

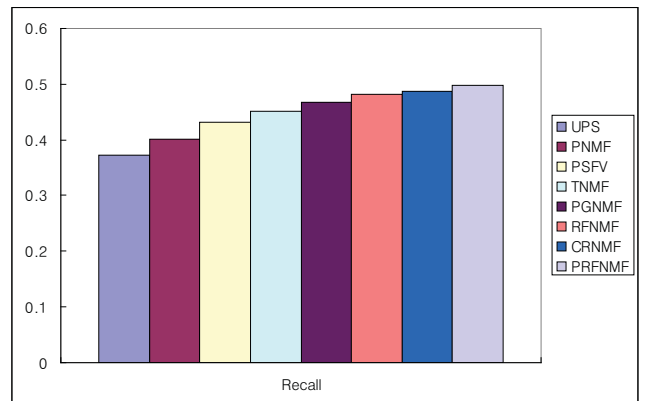


Figure 3. Evaluation Results with respect to average recall

¹ <http://kr.news.yahoo.com> (2009)

The evaluation results are shown in Figure 3. The average recall of PRFNMF is approximately 25.3% higher than that of UPS, 19.4% higher than that of PNMf, 13.3% higher than that of PSFV, 9.3% higher than that of TNMF, 6.2% higher than that of PGNMF, 3.2% higher than that of RFNMF, and 2.2% higher than that of CRNMF.

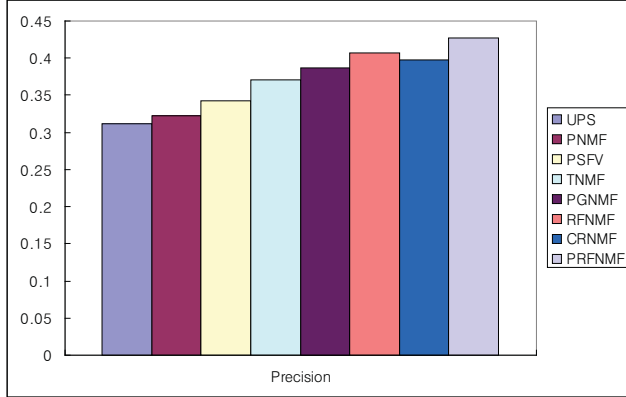


Figure 4. Evaluation Results with respect to average precision

The evaluation results are shown in Figure 4. The average precision of PRFNMF is approximately 27.1% higher than that of UPS, 24.4% higher than that of PNMf, 19.9% higher than that of PSFV, 13.1% higher than that of TNMF, 9.4% higher than that of PGNMF, 4.7% higher than that of RFNMF, and 6.8% higher than that of CRNMF.

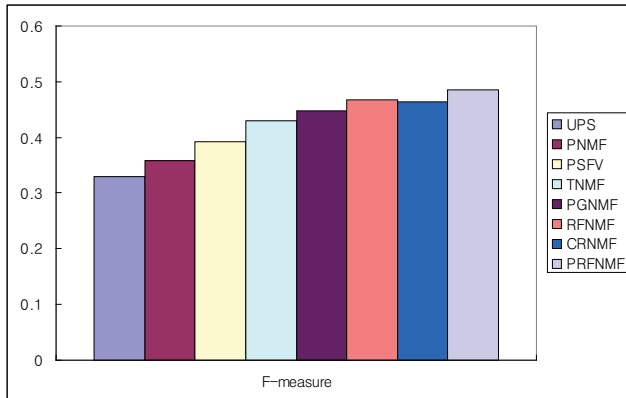


Figure 5. Evaluation Results with respect to average F-measure

The evaluation results are shown in Figure 5. The average F-measure of PRFNMF is approximately 32.2% higher than that of UPS, 26.2% higher than that of PNMf, 19.2% higher than that of PSFV, 11.3% higher than that of TNMF, 7.8% higher than that of PGNMF, 3.7% higher than that of RFNMF, and 4.3% higher than that of CRNMF.

The result shows that recall, precision, and F-measure of PNMf are better than those of the UPS because the PNMf uses semantic features with respect to query-based summary to user's interesting. The result shows that evaluation measures of PSFV are better than

those of the PNMf because the PSFV generates more meaningful summary by reflecting the inherent semantics of document with respect to generic and query-based summary. The result shows that the evaluation measures of TNMF are better than those of the PSFV because the TNMF uses the weighted semantic features. The result shows that the evaluation measures of PGNMF are better than those of the TNMF because the PGNMF uses the weighted semantic features and topic-based sentence ranking. The result shows that the evaluation measures of CRNMF are better than those of the PGNMF because the CRNMF generates meaningful personalized summary by means of the semantic feature and semantic variable reflecting the inherent structure in document for user interesting. The result shows that the evaluation measures of RFNMF are better than those of the CRNMF because the RFNMF generates meaningful personalized summary by using interaction of user interesting. The result shows that the evaluation measures of PRFNMF is best performance because the PRFNMF select meaningful sentences covering the major topics and user interesting of the search results by using the PRF with NMF.

5. Conclusion

A personalized summary adapt to user to correctly identify whether a document is really interesting for him without having to read the whole document. In this paper, we propose the automatic personalized query-based document summarization method using PRF with NMF. The proposed method can select that are highly relevant to a user interesting because it can cover the major and sub-topics of the search results with respect to generic relevance weight and personalized score by using semantic features and semantic variable by means of user interesting. It also can improve the quality of summarization since extracting sentences to reflect the inherent semantics of the search results by generic and query-based summarizations. Finally, it can select sentences that are highly relevant to a given query because it can expand the query with respect to user interesting by using PRF with NMF. Experimental results show that the proposed method outperforms the 7 different summarization methods.

In the near future, we plan to enhance the personalized summarization method by using a variety of weighting terms. We anticipate that it can improve the accuracy of automatic personalized summarization by using cluster refinement.

6. REFERENCES

- [1] Diaz, A., Gervas, P., "User-model based personalized summarization", Information Processing and Management, 43, 2007, pp.1715-1734.
- [2] Goldstein, J., Kantrowitz, M., Mittal, V., Carbonell, J., "Summarizing Text Documents: Sentence Selection and Evaluation Metrics", In proceeding of the ACM SIGIR conference on research and development in information retrieval (1999) 121-128
- [3] Han, K. S., Bea, D. H., Rim, H. C., "Automatic Text Summarization Based on Relevance Feedback with Query Splitting", In proceedings of the 5th International Workshop on Information Retrieval with Asian Language (2000) 201-202

- [4] Kang, S. S., "Information Retrieval and Morpheme Analysis", HongReung Science Publishing Company, 2002.
- [5] Ko, Y. J., An, H. K., Seo, J. Y., "Pseudo-relevance feedback and statistical query expansion for web snippet generation", *Information Processing Letters* 109 (2008) 18-22
- [6] Lee, D. D., Seung, H. S., "Learning the parts of objects by non-negative matrix factorization", *Nature* 401, 1999, pp.788-791.
- [7] Lee, D. D., Seung, H. S., "Algorithms for non-negative matrix factorization", In *Advances in Neural Information Processing Systems*, 13, 2000, pp.556-562.
- [8] Lee, J. H., Park, S., Ahn, C. M., Kim, D. H., "Automatic Generic Document Summarization Based on Non-negative Matrix Factorization", *Information Processing and Management*, 45(1), 2009, pp.20-34.
- [9] Mani, I., "Automatic Summarization", John Benjamins Publishing Company, 2001.
- [10] Park, S. Lee, J. H. Ahn, C. M. Hong, J. S. Chun, S. J., "Query Based Summarization using Non-negative Matrix Factorization", In *proceeding of the International Conference on Knowledge-Based & Intelligent Information & Engineering Systems*, 2006, pp.84-87.
- [11] Park, S., Lee, J. H., "Topic-based Multi-document Summarization Using Non-negative Matrix Factorization and K-means", *Journal of KIISE: Software and Applications*, 35(4), 2008, pp.255-264.
- [12] Park, S., "Generic Summarization using Non-negative Semantic Variable", In *proceeding of ICIC*, 2008.
- [13] Park, S., "Automatic Personalized Summarization using Non-negative Matrix and Relevance Measure", In *proceeding of the IEEE International Workshop on Semantic Computing and Application (IWSCA)*, 2008, pp.72-76.
- [14] Park, S., "Personalized Document Summarization using Non-negative Semantic Feature and Non-negative Semantic Variable", In *proceeding of IDEAL*, 2008.
- [15] Park, S., Lee, J. H., Song, J. W., "Personalized Summarization Agent Using Non-negative Matrix Factorization", In *proceeding of PRICAI*, 2008.
- [16] Park, S. "Document Summarization using Non-negative Matrix Factorization and Relevance Feedback", In *proceeding of ICHIT'08*, 2008.
- [17] Park, S. Char, B. R. "Query based Personalized Summarization Agent using NMF and Relevance Feedback", In *proceeding of ICCIT'08*, 2008.
- [18] Park, S. "Automatic Personalized Multi-document Summarization using NMF with Cluster Refinement", Honam University, Technical Report, 2009.
- [19] Park, S., "Automatic Personalized Text Summarization Agent using Generic Relevance Weight based on NMF", In *proceeding of ICOIN*, 2009.
- [20] Ricardo, B. Y., Berthier, R. N., "Moden Information Retrieval", ACM Press, 1999.
- [21] Sanderson, M., "Accurate user directed summarization from existing tools", In *proceeding of the international conference on information and knowledge management*, 1998, pp.45-51.
- [22] Tombros, A., Sanderson, M., "Advantages of Query Biased Summaries in Information Retrieval", In *proceeding of ACM SIGIR*, 1998, pp.2-10.
- [23] Varadarajan, R., Hristidis, V., "A System for Query-Specific Document Summarization", In *proceeding of the CIKM*, 2006, pp.622-631.
- [24] Wild, S., Curry, J., Dougherty, A., "Motivating Non-Negative Matrix Factorizations", In *proceeding of SIAM ALA*, 2003.